The Invariance Principle — problem 1.48

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In Fig. 1.7, two squares are neighbors if they have a common boundary. Consider the following operation T : Choose any two neighboring numbers and add the same integer to them. Can you transform Fig. 1.7 into Fig. 1.8 by iteration of T ?

1	2	3
4	5	6
7	8	9

Fig. 1.7

7	8	9
6	2	4
3	5	1

Fig. 1.8